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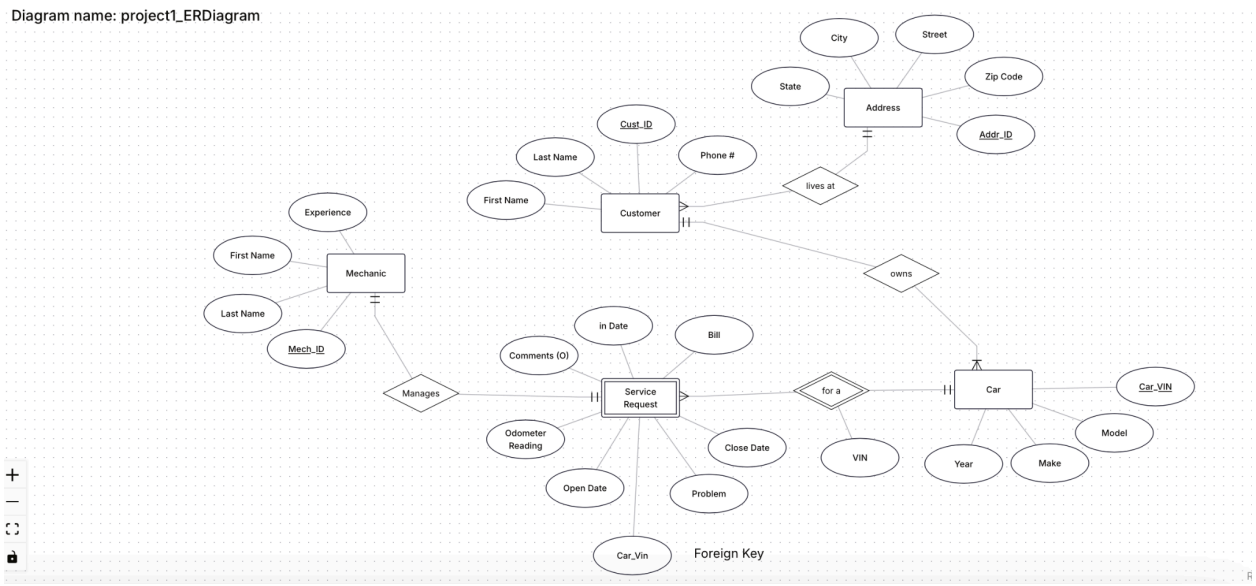
# CS166 Project #1 Phase 2

## Project Overview

We built a database for a mechanics shop. The system tracks information about customers, cars, mechanics, car ownership, service request and billing information.

## ER Diagram & Assumptions




**Entities:**

- Customer:
  - Entity Type: Strong
  - Primary Key: Cust_ID
  - Cardinality & Participation Constraints:
    - Optional
      - None
    - Mandatory
      - A Customer **<u>must</u>** own at least 1 to many cars (1-to-many)
      - A Customer **<u>must</u>** live at at most only one address (1-to-1)

- Address:

    - Entity Type: Strong
    - Primary Key: Addr_ID
    - Cardinality & Participation Constraints:
        - Optional
            - An Address **can** house 0 to many customers (1-to-many)
        - Mandatory
            - None

- Car:
    - Entity Type: Strong
    - Primary Key: VIN
    - Cardinality & Participation Constraints:
        - Optional:
            - Car **can** have 0 or many outstanding service requests (1-to-many)
        - Mandatory:
            - Car **must** have at most one owner (1-to-1)

- Mechanic:
    - Entity Type: Strong
    - Primary Key: Mech_ID
    - Cardinality & Participation Constraints:
        - Optional:
            - None
        - Mandatory:
            - Mechanic **may** have worked on one or more service requests (1-to-many)

- Service Request:
    - Entity Type: Strong
    - Super key: Req_id
    - Cardinality & Participation Constraints:
        - Mandatory:
            - A service request **must** have at most one Mechanic working on it at a time and many closed service requests can belong to one mechanic (many-to-1)
            - A Service Request **must** be made for at most one car (1-to-1)

**Relationships:**

- [Customer] Lives at [Address]
    - Relationship Type: Weak
    - Attribute: None

- [Mechanic] Manages [Service Request]
    - Relationship Type: Strong
    - Attribute: None

- [Service Request] For a [Car]
    - Relationship Type: Strong
    - Attribute: VIN

- [Customer] owns a [Car]
    - Relationship Type: Strong
    - Attribute: None

## Client Application Features

**Query #1: Add Customer**

- Test Case:
    - Id: 22222
    - First name: Pete
    - Last name: Smith
    - Phone: 1234567890
    - Addr id: 10001
- Screenshot:

```
[vzamb003@xe-10 project_1]$ source compile.sh


*******************************************************
              R&V Mechanic Shop Database
*******************************************************

Connecting to database...Connection URL: jdbc:postgresql://localhost:20687/vzamb003_proj_1_DB

Done
MAIN MENU
---------
1. Add Customer
2. Add Mechanic
3. Add Car
4. Start Service Request
5. Close Service Request
6. List date, comment, and bill for all closed requests with bill lower than 100
7. List First Name and Last Name of Customers having more than 20 different cars
8. List Make, Model, and Year of all cars build before 1995 having less than 50000 miles
9. List the Make, Model and number of Service Requests for the first k Cars with the highest number of Service Requests
10. List the first name, last name and total bill of customers in descending order of their total bill for all cars brought to the mechanic.
11. Update the mechanic for an existing service request
12. List all available mechanics (i.e., mechanics that are not currently working on a service request)
13. List all open service requests
14. List the average repair costs per car make
15. List all customers with an open service request
16. < EXIT
Welcome to R&V Mechanic Shop Database! Please make a selection of the following queries: 1
        Enter 5-digit Customer ID: 22222
        Enter First Name: Pete
        Enter Last Name: Smith
        Enter 10-digit Phone Number: 1234567890
        Enter 5-digit Address ID: 10001
addr_id
10001
Customer added successfully!
MAIN MENU
```

**Query #2: Add Mechanic**

- Test Case:
    - Id: 22222
    - First name: Pete
    - Last name: Smith
    - Phone: 1234567890
    - Addr id: 10001
- Screenshot:

```
MAIN MENU
---------
1. Add Customer
2. Add Mechanic
3. Add Car
4. Start Service Request
5. Close Service Request
6. List date, comment, and bill for all closed requests with bill lower than 100
7. List First Name and Last Name of Customers having more than 20 different cars
8. List Make, Model, and Year of all cars build before 1995 having less than 50000 miles
9. List the Make, Model and number of Service Requests for the first k Cars with the highest number of Service Requests
10. List the first name, last name and total bill of customers in descending order of their total bill for all cars brought to the mechanic.
11. Update the mechanic for an existing service request
12. List all available mechanics (i.e., mechanics that are not currently working on a service request)
13. List all open service requests
14. List the average repair costs per car make
15. List all customers with an open service request
16. < EXIT
Welcome to R&V Mechanic Shop Database! Please make a selection of the following queries: 2
        Enter 5-digit Mechanic ID: 33333
        Enter First Name: John
        Enter Last Name: Doe
        Enter Years of Experience (0-99): 20
Mechanic added successfully!
MAIN MENU
```

**Query #3: Add Car**

- Test Case:
    - VIN: 99999
    - Year: 2018
    - Make: Toyota
    - Model: Corolla
    - Customer: 20001
- Screenshot:

```
[vzamb003@xe-10 project_1]$ source compile.sh

Connecting to database...Connection URL: jdbc:postgresql://localhost:20687/vzamb003_proj_1_DB

Done
MAIN MENU
---------
1. Add Customer
2. Add Mechanic
3. Add Car
4. Start Service Request
5. Close Service Request
6. List date, comment, and bill for all closed requests with bill lower than 100
7. List First Name and Last Name of Customers having more than 20 different cars
8. List Make, Model, and Year of all cars build before 1995 having less than 50000 miles
9. List the Make, Model and number of Service Requests for the first k Cars with the highest number of Service Requests
10. List the first name, last name and total bill of customers in descending order of their total bill for all cars brought to the mechanic.
11. Update the mechanic for an existing service request
12. List all available mechanics (i.e., mechanics that are not currently working on a service request)
13. List all open service requests
14. List the average repair costs per car make
15. List all customers with an open service request
16. < EXIT
Welcome to R&V Mechanic Shop Database! Please make a selection of the following queries: 3
        Enter 5-digit VIN number: 99999
        Enter year of vehicle: 2018
        Enter make: Toyota
        Enter model: Corolla
        Enter 5-digit customer ID of owner: 20001
cust_id
20001
20001
20001
20001
Car added successfully!
```

Screenshot

**Query #4: Start Service Request**

- Test Case:
  - Customer last name: Smith
  - Customer ID: 20001
  - VIN: 90002
  - Mechanic ID: 30002
  - Request ID: 50014
  - In Date: 260308
  - Problem: Paint Chipping
  - Start Date: 260309
  - Close Date: EMPTY
  - Comments: In Progress
  - Bill: 500
  - Odometer: 12000
- Screenshot:

```
[vzamb003@xe-10 project_1]$ source compile.sh
Welcome to R&V Mechanic Shop Database! Please make a selection of the following queries: 4
        Enter last name of the Customer of the Service Request: Smith
lastname
Smith
Smith
cust_id firstname       lastname
20001   John    Smith
22222   Pete    Smith
        Multiple Customers found. Select a Customer ID from the list above: 20001
vin     year    make    model
90001   2018    Toyota  Camry
90002   2022    Tesla   Model 3
30001   2018    Toyota  Camry
30006   2016    Toyota  Corolla
99999   2018    Toyota  Corolla
        Select the VIN of the car for the Service Request: 90002

Available Mechanics:
mech_id firstname       lastname        experience
30002   Elena   Gears   5
30003   Mike    Wrencher        8
40001   Mike    Garcia  10
40002   Anna    Lopez   6
40006   Daniel  White   7
40007   Sophia  Clark   5
33333   John    Doe     20
        Enter Mechanic ID from the list above: 30002
Enter Service Request ID: 50014
Enter the date car was brought in (yymmdd): 260308
Enter the reason for visit: Paint Chipping
If applicable, Enter start date (yymmd) or press ENTER if none: 260309
If applicable, Enter close date (yymmd) or press ENTER if none:
Enter comments or press ENTER if none: In Progress
Enter bill amount: 500
Enter odometer reading: 12000

Service Request added successfully!
MAIN MENU
```

**Query #5: Close Service Request**

- Test Case:
  - Service Request #: 50013
  - Mechanic ID: 30001
  - Close Date: 260310
  - Comments: [ENTER - error check], Done
  - Total Bill: 600
- Screenshot:

```
[vzamb003@xe-10 project_1]$ source compile.sh
MAIN MENU
---------
1. Add Customer
2. Add Mechanic
3. Add Car
4. Start Service Request
5. Close Service Request
6. List date, comment, and bill for all closed requests with bill lower than 100
7. List First Name and Last Name of Customers having more than 20 different cars
8. List Make, Model, and Year of all cars build before 1995 having less than 50000 miles
9. List the Make, Model and number of Service Requests for the first k Cars with the highest number of Service Requests
10. List the first name, last name and total bill of customers in descending order of their total bill for all cars brought to the mechanic.
11. Update the mechanic for an existing service request
12. List all available mechanics (i.e., mechanics that are not currently working on a service request)
13. List all open service requests
14. List the average repair costs per car make
15. List all customers with an open service request
16. < EXIT
Welcome to R&V Mechanic Shop Database! Please make a selection of the following queries: 5
        Enter service request number: 50013
req_id
50013
        Enter mechanic ID: 30001
        Enter Service Request's close date (yymmdd): 260310
        If applicable, enter any additional comments. Must include 'Done' before enter:
        Comments must include the word 'Done' before closing the request.
        If applicable, enter any additional comments. Must include 'Done' before enter: Done
        Enter total bill amount: 600
Service Request closed successfully!
MAIN MENU
```

**Query #6: List date, comment, and bill for all closed requests with bill lower than 100**

- Test Case:
  - No Input needed
- Screenshot:

```
Welcome to R&V Mechanic Shop Database! Please make a selection of the following queries: 6
closedate       comments        bill
240105  Done Routine oil change 89.99
240125  Done Rotated tires      45.00
11026   Done    85.00
total row(s): 3
```

**Query #7: List First Name and Last Name of Customers having more than 20 different cars**

- Test Case:
  - No input needed
- Screenshot (not enough data in our .txt files to run the query):

```
Welcome to R&V Mechanic Shop Database! Please make a selection of the following queries: 7
total row(s): 0
```

**Query #8: List Make, Model, and Year of all cars build before 1995 having less than 50000 miles**

- Test Case:
  - No input needed
- Screenshot:

```
Welcome to R&V Mechanic Shop Database! Please make a selection of the following queries: 8
make    model   year
Honda   PRelude 1992
total row(s): 1
```

**Query #9: List the Make, Model and number of Service Requests for the first k Cars with the highest number of Service Requests**

- Test Case:
  - k = 3
- Screenshot:

```
Welcome to R&V Mechanic Shop Database! Please make a selection of the following queries: 9
        Enter k (positive integer): 3
make    model   num_service_requests
Nissan  Altima  2
Tesla   Model 3 2
Honda   Civic   2
total row(s): 3
```

**Query #10: List the first name, last name and total bill of customers in descending order of their total bill for all cars brought to the mechanic.**

- Test Case:
  - No input needed
- Screenshot:

```
Welcome to R&V Mechanic Shop Database! Please make a selection of the following queries: 10
cust_id firstname       lastname        customertotalbill
20003   Michael Brown   1860.00
20001   John    Smith   1716.25
20005   David   Wilson  640.00
20002   Emily   Johnson 455.49
20004   Sarah   Davis   120.00
total row(s): 5
```

**EXTRA: Query #11: Update the mechanic for an existing service request**

- Test Case:
  - Service Req ID: 50008
  - Mechanic ID: 40002
- Screenshot:

```
Welcome to R&V Mechanic Shop Database! Please make a selection of the following queries: 11
Enter 5-digit service request id
50008
req_id
50008
Enter 5-digit mechanic id
40002
mech_id
40002
Service Request's Mechanic Updated Successfully!
MAIN MENU
```

**EXTRA: Query #12: List all available mechanics (i.e., mechanics that are not currently working on a service request**

- Test Case:
    - No input needed
- Screenshot:

```
Welcome to R&V Mechanic Shop Database! Please make a selection of the following queries: 12
mech_id firstname       lastname        experience
30001   Dave    Fixit   12
30003   Mike    Wrencher        8
40003   Chris   Martinez        12
40004   Kevin   Lee     4
40006   Daniel  White   7
33333   John    Doe     20
total row(s): 6
MAIN MENU
```

**EXTRA: Query #13: List all open service requests**

- Test Case:
    - No input Needed
- Screenshot:

```
Welcome to R&V Mechanic Shop Database! Please make a selection of the following queries: 13
req_id  vin     mech_id opendate        problem
50005   30005   40005   240115  Transmission slip
50014   90002   30002   260309  Paint Chipping
50015   90099   40001   null    Broken Window
50004   30004   40007   240112  Battery issue
50008   30003   40002   240128  Check engine light
total row(s): 5
```

**EXTRA: Query #14: List the average repair costs per car make**

- Test Case:
    - No input needed
- Screenshot:

```
Welcome to R&V Mechanic Shop Database! Please make a selection of the following queries: 14
make    avg_repair_cost
Ford    440.2500000000000000
Honda   478.3300000000000000
Tesla   406.6666666666666667
Toyota  205.4166666666666667
Nissan  640.0000000000000000
total row(s): 5
```

**EXTRA: Query #15: List all customers with an open service request**

- Test Case:
    - No input needed
- Screenshot:

```
Welcome to R&V Mechanic Shop Database! Please make a selection of the following queries: 15
firstname       lastname        req_id  make    model
David   Wilson  50005   Nissan  Altima
John    Smith   50014   Tesla   Model 3
Michael Brown   50015   Honda   PRelude
Sarah   Davis   50004   Tesla   Model3
Michael Brown   50008   Ford    Focus
total row(s): 5
```